EXPERIMENT 20

AIM:

To implement the K-Means clustering algorithm on a given dataset.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('Cafe_Customers_KMeans.csv')

# Display dataset information
df.info()

# Display first five rows
print(df.head())

# Visualize the dataset
sns.pairplot(df)
plt.show()

# Select features for clustering
features = df.iloc[:, [1, 2]].values
```

```
# Apply K-Means clustering
from sklearn.cluster import KMeans

model = KMeans(n_clusters=3, random_state=42, n_init=10)
model.fit(features)

# Create final DataFrame with cluster labels
Final = df.iloc[:, [1, 2]].copy()
Final['label'] = model.predict(features)

print(Final.head())

# Visualize the clusters
sns.set_style("whitegrid")

sns.FacetGrid(Final, hue='label', height=6) \
    .map(plt.scatter, 'MonthlyVisits', 'AverageSpend') \
    .add_legend()

plt.show()

# Elbow method to find the optimal number of clusters
features_el = df.iloc[:, [1, 2]].values
```

```
wcss = []

for i in range(1, 10):
    model = KMeans(
        n_clusters=i,
        random_state=42,
        n_init=10
    )
    model.fit(features_el)
    wcss.append(model.inertia_)

# Plot the Elbow graph
plt.plot(range(1, 10), wcss, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()

# Final K-Means model
finalModel = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)
```

```
finalModel.fit(features)


# Assign final cluster labels
Final['label'] = finalModel.predict(features)


print(Final)
```

OUTPUT:

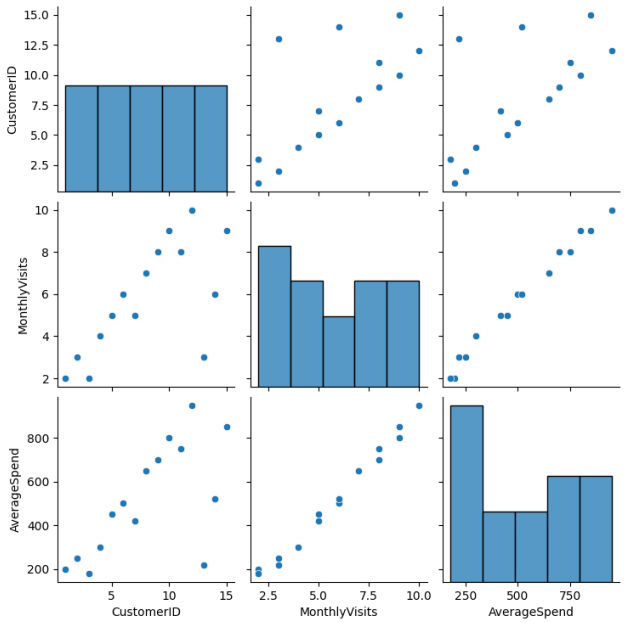
1. Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 3 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   CustomerID     15 non-null     int64
 1   MonthlyVisits  15 non-null     int64
 2   AverageSpend   15 non-null     int64
dtypes: int64(3)
memory usage: 492.0 bytes
```

2. Original Dataset – First Five Records

	CustomerID	MonthlyVisits	AverageSpend
0	1	2	200
1	2	3	250
2	3	2	180
3	4	4	300
4	5	5	450

3. Pair Plot



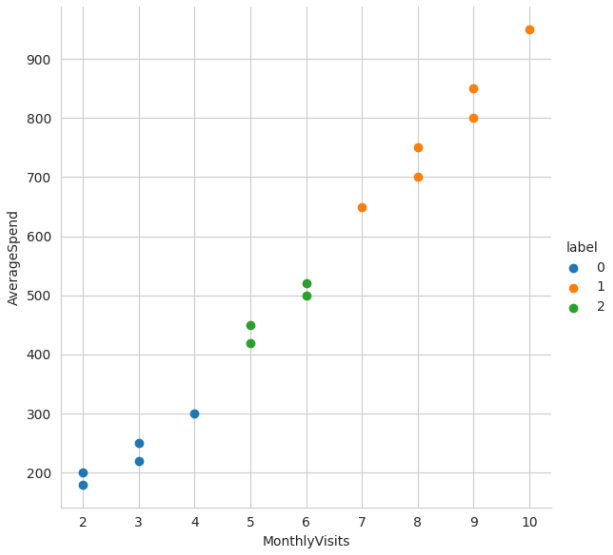
4. K-Means Model

```
KMeans(n_clusters=3, n_init=10, random_state=42)
```

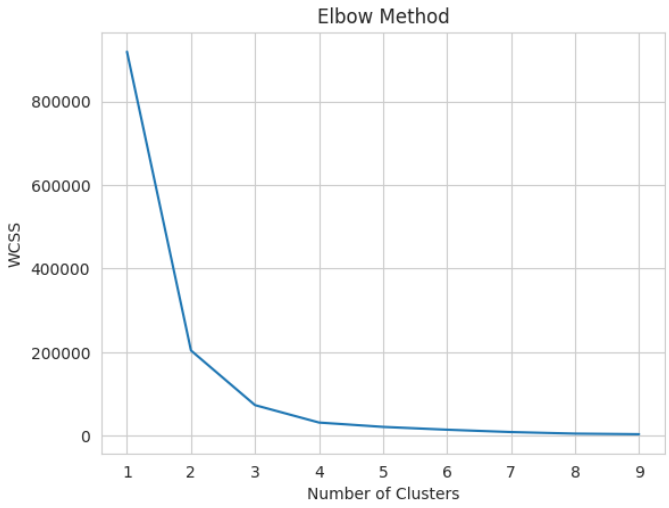
5. Clustered Data – First Five Records

	MonthlyVisits	AverageSpend	label
0	2	200	0
1	3	250	0
2	2	180	0
3	4	300	0
4	5	450	2

6. Cluster Visualization



7. Elbow Method – WCSS



8. Final K-Means Model

```
KMeans(n_clusters=3, n_init=10, random_state=42)
```

9. Final Clustered Dataset

	MonthlyVisits	AverageSpend	label
0	2	200	0
1	3	250	0
2	2	180	0
3	4	300	0
4	5	450	2
5	6	500	2
6	5	420	2
7	7	650	1
8	8	700	1
9	9	800	1
10	8	750	1
11	10	950	1
12	3	220	0
13	6	520	2
14	9	850	1

10. Final Cluster Output

```
Customer groups are represented by the labels 0, 1 and 2.
The final dataset shows the cluster assigned to each customer
based on
MonthlyVisits and AverageSpend.
```

RESULT:

The café customers were successfully grouped into clusters using K-Means clustering based on their monthly visits and average spending.